MicroTrade Pro

Sunlight, traded next door.

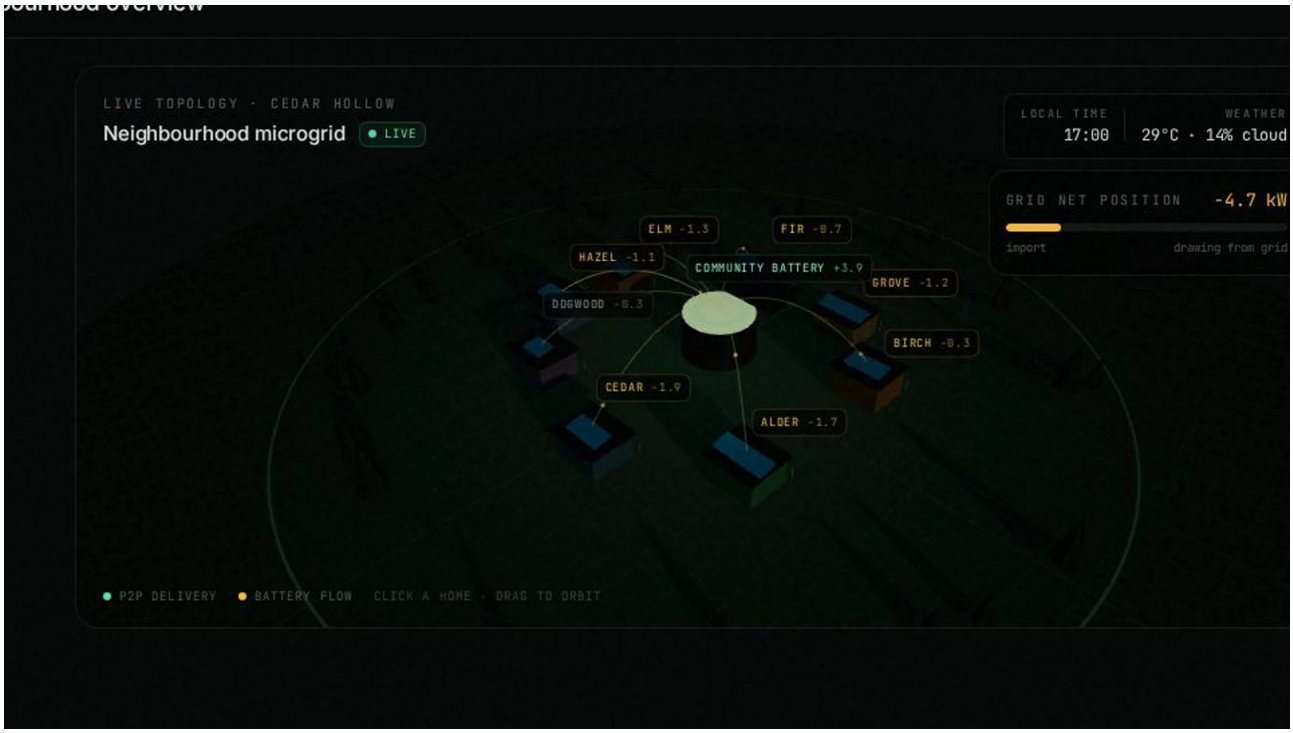
Predictive peer-to-peer energy trading · IEEE ClimateChain 2026, Track 2

The problem

Most rooftop solar is wasted -- exported for pennies while the evening peak is bought back at full price. The grid takes its cut both ways, and the carbon savings are never counted.

What it does

- **Predict.** Each of 9 household agents trains its own ridge-ensemble forecaster on three weeks of meter history plus live weather: 48-hour load and solar curves with P10-P90 bands, ~10% error, 62% better than guessing.
- **Trade.** Every morning the agents bid into a uniform-price double auction -- one clearing price per hour. Matches become forward contracts: buyer escrows payment, seller stakes a 12% bond.
- **Settle.** Hourly delivery is metered; escrow releases, shortfalls slash bonds, and each delivered green kWh mints a GreenREC. Every write is a signed, Merkle-committed ledger transaction -- mirrored on-chain.



Live 3D microgrid: nine homes around a community battery

Smart contracts -- deployed, not vaporware

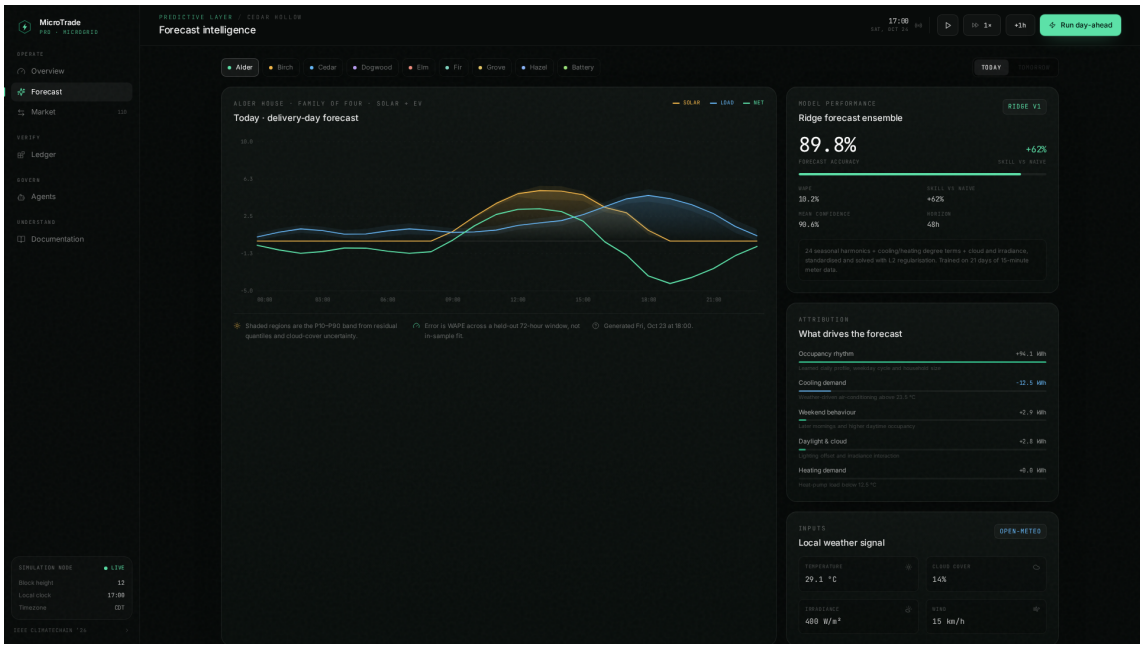
V2 is the hardened build: reentrancy guard, staked seller bonds, oracle-gated settlement and minting, zeroed balances before external calls. Sources in contracts/ (MIT).

Base Sepolia (84532, headline): `0x45C0Be734cF808fcab42E285F33a7a811cc8EB00`

Ethereum Sepolia V2 mirror: `0x2f7EbFA1942B97F74D428226CcA48dD643794E78`

Ethereum Sepolia V1: `0x8d1e4ad61a2e4071707b512473652924b8E4c914`

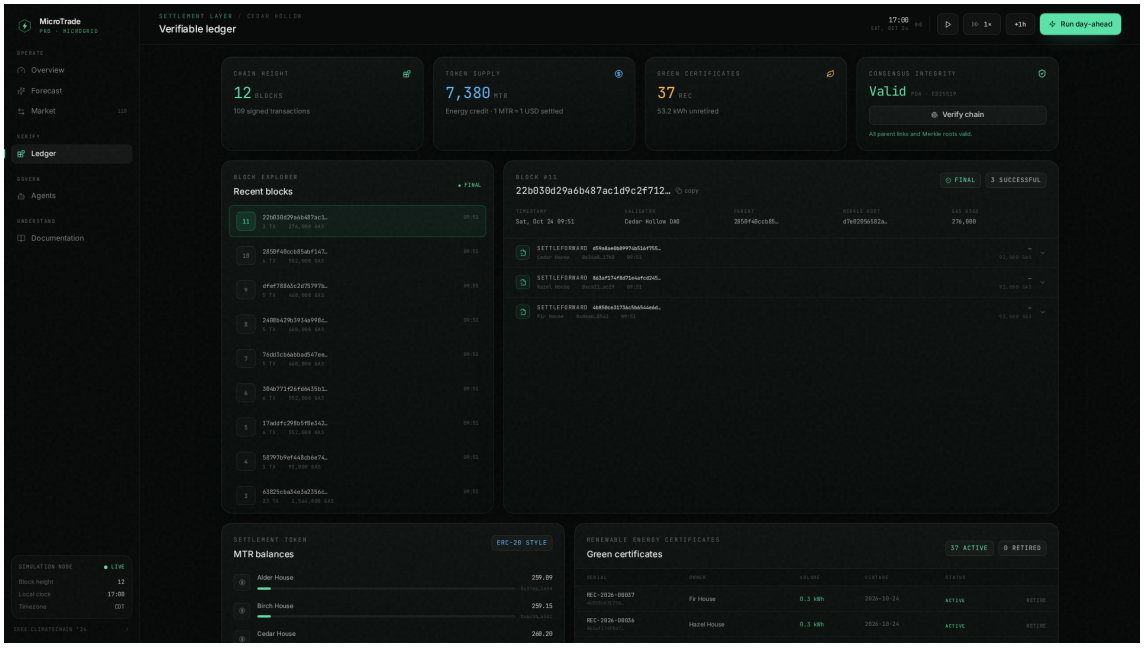
Honest scope: testnet deployments, self-reviewed and hardened -- not a third-party audit. The code is open and every claim is checkable on-chain.



48-hour forecast with uncertainty bands



Day-ahead double auction and forward book



How it was built

React + Three.js live microgrid, Node/Express deterministic simulation engine, dependency-free ridge regression, Ed25519-signed ledger, Solidity V2 on Base Sepolia. Demo film cut with Hyperframes from real app capture, voiced with Deepgram.

What's next

Mainnet pilot with a real community battery, oracle feeds for settlement, third-party audit of V2, and smart-plug metering integrations.

Links

Live demo: https://microtrade-pro.onrender.com/

Code (MIT): https://github.com/ThatLinuxGuyYouKnow/microtrade-pro

Demo video: YouTube link in the Devpost entry